

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1031

Vergennes 9 138 kV
August 25, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1031. The Interconnection Customer proposes a 56.8 MW solar photovoltaic facility interconnecting at the Vergennes 9 138 kV substation in the West Virginia area. The request was received on July 21, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 266 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,880,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Heyburn to Lysander 138 kV	Sherbino I to Coal 138 kV	134.7	210	6.5
Raver to Sherman Rd 138 kV L	Coal to Sherino I 138 kV L	132.6	224	6.5
Raver to Lsander 138 kV	Sherbino I Substation to Coal 138 kV line	124.2	236	6.5
Pawnee to Alpha Rd 138 kV	Beech Ridge Energy Llc 138 kV/1 kV Autotransformer	122.6	193	10.5
Bradley to Phelan 138 kV	Scarbro to N Swetwater Substation 138 kV	117.0	232	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Heyburn to Lysander 138 kV	Rebuild line with higher-capacity conductor	\$18,748,000	9% (\$1,694,000)
Raver to Sherman Rd 138 kV L	Rebuild line with higher-capacity conductor	\$5,505,000	9% (\$498,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Raver to Lysander 138 kV	Rebuild line with higher-capacity conductor	\$4,412,000	9% (\$399,000)
Alpha Road to Pawnee Substation 138 kV	Rebuild line with higher-capacity conductor	\$6,023,000	22% (\$1,340,000)
Bradley to Phelan 138 kV	Rebuild line with higher-capacity conductor	\$18,393,000	49% (\$8,949,000)
	Total allocated to project		\$12,880,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.